

Notes: MacKay and Phillips (RFS, 2005)

How Does Industry Affect Firm Financial Structure?

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1 Big picture

- **Question.** How important is **industry** for firm capital structure once we move beyond standard industry fixed effects, and can a firm's **position within its industry** explain why firms in the same industry look so different along leverage, technology, and risk dimensions?
- **Setting.** The paper studies **U.S. manufacturing firms** from the merged **COMPUSTAT–CRSP** panel over **1981–2000**, and compares firms in **competitive industries** to firms in **concentrated industries**.
- **Core challenge.** Standard capital-structure work often uses industry dummies or industry averages, but that does not explain the large **within-industry dispersion** in leverage. The paper asks whether this dispersion reflects **industry equilibrium forces** rather than idiosyncratic noise.
- **Main contribution.** The paper connects corporate finance to **industry equilibrium models**. It shows that financial structure depends not only on broad industry membership but also on:
 1. a firm's **natural hedge** (how close its capital-labor ratio is to the industry median),
 2. the **actions of peer firms** in the same industry,
 3. and, more weakly, whether the firm is an **entrant, incumbent, switcher, seller, or exiter**.
- **Why it matters.** If leverage, technology, and risk are jointly determined inside industries, then empirical work that treats financing as an isolated choice can be misleading. The paper argues that **real and financial decisions must be studied together**.
- **Main empirical question.** In competitive industries, do firms close to the technological core of the industry carry less leverage, and do firms adjust in response to what their peers do? Are those patterns weaker or stronger in concentrated industries?
- **Main empirical results.**
 - **Industry fixed effects explain only a small share** of leverage variation: about **13%**, while firm fixed effects explain **54%**, with the remaining **33%** coming from within-firm variation over time.
 - Firms in **competitive industries** that are closer to the **industry median capital-labor ratio** use **less financial leverage**.
 - Changes in a firm's leverage, capital intensity, and risk are **inversely related** to changes made by other firms in the same industry.
 - In **concentrated industries**, leverage is **higher and less dispersed**, the natural-hedge effect largely disappears, and peer interactions become **more important**.
 - The data support several predictions of the **static competitive-industry equilibrium** model of Maksimovic and Zechner (1991), but offer only **limited support** for the more elaborate dynamic entry–exit models of Williams (1995) and Fries, Miller, and Perraudin (1997).
 - Estimating leverage, capital intensity, and risk **simultaneously** matters economically, not just econometrically.

- **Economic takeaway.** A firm’s capital structure is partly an **industry-relative object**. Departures from industry leverage norms tend to coincide with departures from industry technology and risk norms.

2 Data and measurement (key objects)

2.1 Data sources and sample

- **Primary data.** The paper uses the merged **COMPUSTAT–CRSP** database produced by WRDS.
- **Financial and operating variables.** These come from **COMPUSTAT**.
- **Historical industry classifications.** These come from **CRSP**, because COMPUSTAT reports only current industry codes.
- **Diversification data.** The authors merge COMPUSTAT segment files to construct output Herfindahls.
- **Industry concentration.** The paper uses four-digit SIC **Herfindahl–Hirschman Index (HHI)** measures from the **Census of Manufacturers** for 1982, 1987, 1992, and 1997.
- **Reason for deletion.** COMPUSTAT’s deletion code is used to distinguish true firm exit (bankruptcy or liquidation) from corporate restructuring.

2.2 Competitive versus concentrated industries

- **Competitive industries:** $HHI < 1000$.
- **Concentrated industries:** $HHI > 1800$.
- Industries with HHI between 1000 and 1800 are dropped to sharpen the comparison.
- The sample is limited to **manufacturing industries** (roughly SIC 2000–3990), excluding miscellaneous 99-ending industries.

2.3 Final sample

- Because the multivariate regressions use **first differences** and **second lags** as instruments, the final usable panel is smaller than the raw database.
- **Competitive-industry sample:** 3,074 firms and 17,140 firm-years in 315 industries.
- **Concentrated-industry sample:** 309 firms and 1,630 firm-years in 46 industries.
- The panel is **unbalanced**.

2.4 Key firm-level variables

- **Financial leverage.** Total debt divided by total assets (book leverage in the baseline).

- **Capital intensity.** Fixed capital stock (net PP&E) divided by number of employees:

$$\text{Capital/Labor}_{ft} = \frac{\text{Net PP\&E}_{ft}}{\text{Employees}_{ft}}.$$

- **Risk.** Standard deviation of operating cash flow divided by total assets, using up to 10 annual observations and at least 4.
- **Profitability.** EBIT divided by total assets.
- **Diversification.** One minus the Herfindahl of output across the firm's four-digit SIC segments.
- **Tobin's q .** Market value of equity plus book debt and preferred stock minus deferred taxes, divided by book assets.

2.5 Industry-position variables

- The paper's core idea is that what matters is not just *which* industry a firm is in, but *where* it sits inside that industry.
- **Natural hedge.** This measures how close a firm's capital-labor ratio is to the market-share-weighted industry-year median capital-labor ratio. It is scaled to lie in $[0, 1]$, where 1 means the firm sits exactly at the industry median and 0 means it is furthest away:

$$\text{NH}_{fiy} = 1 - \frac{|(K/L)_{fiy} - \text{median}_{iy-f}(K/L)|}{\text{range}\{(K/L)_{fiy} - \text{median}_{iy-f}(K/L) \mid \forall f \in i, y\}}.$$

- **Interpretation.** A high natural-hedge firm is technologically close to the **industry core**; a low natural-hedge firm is at the **fringe**.
- **Peer-action measures.** For each dependent variable, the paper defines:
 - **intra-quantile change:** average change among firms in the same half of the industry distribution,
 - **extra-quantile change:** average change among firms in the other half.

These are built from lagged industry rankings and exclude the focal firm itself.

2.6 Entry, switching, selling, and exit

- **Entrants:** firms appearing in 1991–2000 but not in 1981–1990.
- **Switchers:** firms that change two-digit SIC industry across subperiods.
- **Exiters:** firms leaving the sample through Chapter 11 or Chapter 7.
- **Sellers:** firms leaving an industry via mergers, acquisitions, or restructuring rather than genuine failure.
- **Incumbents:** firms observed in both subperiods.

2.7 Key descriptive facts

- **Leverage is more dispersed in competitive industries.** Competitive industries display broader within-industry dispersion in leverage.
- **Leverage is higher in concentrated industries.** Mean leverage is roughly **0.239** in competitive industries and **0.275** in concentrated industries; medians are about **0.212** and **0.256**.
- **Capital intensity is highly industry-specific.** Industry fixed effects explain much more of the variation in capital-labor ratios than in leverage.
- **Entry and exit patterns look like core versus fringe.** Entrants tend to be more leveraged, less capital intensive, and less profitable than incumbents; exiters are highly leveraged, risky, and unprofitable.
- **Firm rankings are persistent.** Large, capital-intensive, profitable firms tend to stay near the top of industry distributions over time, while small and risky firms move in and out.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical background: industry equilibrium rather than representative-firm finance

- The paper builds on three competitive-industry equilibrium papers:
 1. **Maksimovic and Zechner (1991):** firms choose debt and technology jointly; firms near the technological center of the industry enjoy a **natural hedge** and use less debt.
 2. **Williams (1995):** industries have a capital-intensive core and a labor-intensive fringe; entry and exit matter.
 3. **Fries, Miller, and Perraudin (1997):** debt choice evolves in dynamic industry equilibrium with entry and exit.
- The central prediction is that leverage, technology, and risk are **jointly determined within industries**.

3.2 Natural-hedge equation

The paper's main measure of industry position is

$$NH_{fiy} = 1 - \frac{|(K/L)_{fiy} - \text{median}_{iy-f}(K/L)|}{\text{range}\{(K/L)_{fiy} - \text{median}_{iy-f}(K/L) \forall f \in i, y\}}.$$

- **Interpretation.** When NH is high, the firm uses a technology close to the industry norm, so it is naturally hedged and should rely less on debt.
- The paper also includes NH^2 because the data show a non-monotonic relation: the combined effect is still negative overall, but the relation is not purely linear.

3.3 Simultaneous-equation system

The empirical core is a three-equation system in first differences:

$$\Delta\text{Leverage} = f(\Delta\text{Capital/Labor}, \Delta\text{Risk}; \Delta\text{industry position}, \Delta\text{controls}, \text{fixed effects}) + \varepsilon_1,$$

$$\Delta\text{Capital/Labor} = g(\Delta\text{Leverage}, \Delta\text{Risk}; \Delta\text{industry position}, \Delta\text{controls}, \text{fixed effects}) + \varepsilon_2,$$

$$\Delta\text{Risk} = h(\Delta\text{Leverage}, \Delta\text{Capital/Labor}; \Delta\text{industry position}, \Delta\text{controls}, \text{fixed effects}) + \varepsilon_3.$$

- **Dependent variables.** Financial leverage, capital-labor ratio, and cash-flow volatility.
- **Industry-position regressors.** Natural hedge, intra-quantile change, extra-quantile change, and entry/exit dummies.
- **Controls.** Profitability, firm size, diversification, and Tobin's q .
- **Interpretation.** The system operationalizes the idea that financing, technology, and risk choices are made **jointly**, not sequentially.

3.4 Why first differences and GMM?

- The paper uses **first differences** to remove firm fixed effects.
- It uses **industry-year interacted fixed effects** to isolate within-industry variation.
- It instruments endogenous regressors using their **second lags in levels**.
- GMM therefore addresses both:
 1. **firm heterogeneity**, and
 2. **simultaneity / endogeneity bias**.
- The authors also report **OLS** for comparison, and several key signs change between OLS and GMM, showing that simultaneity is not a minor issue.

3.5 Interpretation of the main coefficients

- **Natural hedge in leverage equation.** Negative overall effect in competitive industries. Firms closer to the median technology carry less debt.
- **Intra- and extra-quantile changes.** Negative coefficients across equations. A firm's leverage, capital intensity, and risk tend to move *against* changes made by peer firms, especially peers with similar characteristics.
- **Simultaneity.** The sign on capital intensity in the leverage equation changes between OLS and GMM, and several other coefficients move as well; this is evidence that estimating one equation in isolation is misleading.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- Industry dummies alone capture only broad between-industry effects.
- But the paper documents that most leverage variation is **within industries**, not across them.
- Therefore, identification has to come from **within-industry, within-firm, and within-industry-position** variation.

4.2 What identifies the paper's main effects

- **Firm fixed effects** are removed by first differencing.
- **Within-industry variation** is isolated with interacted industry-year fixed effects.
- **Simultaneity** is addressed using GMM with second-lag instruments.
- **Industry position** is measured in economically meaningful ways: natural hedge, peer changes, and entry/exit status.

4.3 Instrument validity and relevance

- The paper uses Hansen's J -test to assess overidentifying restrictions.
- Except for the risk equation, the overidentifying restrictions are not rejected at conventional levels.
- Instrument relevance, using Shea's procedure, is about:
 - 8% for leverage,
 - 10% for capital-labor,
 - 18% for risk.

4.4 Why the concentrated-industry comparison matters

- The concentrated-industry sample is used as a benchmark.
- If the competitive-industry variables were equally strong there, the results would not be very specific to competitive-industry equilibrium models.
- Instead, the paper finds that **natural hedge is not significant** in concentrated industries, while peer-interaction variables remain strong and become even more economically important.

4.5 Robustness checks

- Results are similar when leverage is measured with **market values** rather than book values.
- Results broadly survive when the sample is restricted to **single-segment firms**.

- Four-digit SIC appears to be the preferred level of industry definition: broader SIC definitions weaken the peer-change results and perform worse econometrically.

5 Tables and figures

5.1 Figure 1: dispersion in leverage for competitive and concentrated industries

- The figure groups industries by how dispersed leverage is within the industry, then plots leverage quintiles inside each group.
- **Read:** there is substantial **within-industry dispersion** even in the least dispersed industries.
- Competitive industries display broader leverage dispersion, while concentrated industries cluster around **higher leverage levels**.
- This figure motivates the entire paper: industry dummies alone cannot summarize what is happening inside industries.

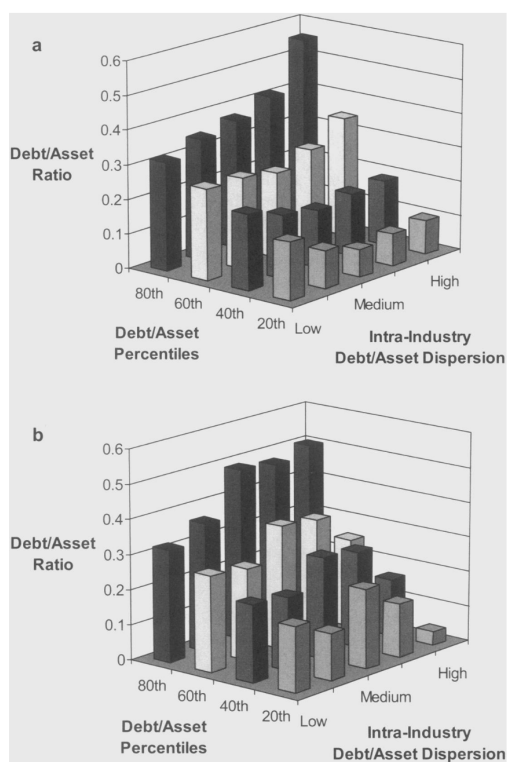


Figure 1
(a) Dispersion in financial leverage for competitive industries. (b) Dispersion in financial leverage for concentrated industries.

Table 1
Summary statistics

	Firm-years	Mean	Median	Standard deviation	Robust dispersion	Range (maximum–minimum)
Competitive industries						
Leverage	17,140	0.239	0.212	0.194	0.203	1.000
		<i>-0.005</i>	<i>-0.028</i>	<i>0.176</i>	<i>0.165</i>	<i>1.427</i>
Capital/Labor	17,140	0.047	0.024	0.085	0.022	1.375
		<i>-0.004</i>	<i>-0.004</i>	<i>0.078</i>	<i>0.015</i>	<i>4.164</i>
Risk	17,140	0.091	0.062	0.089	0.049	0.878
		<i>-0.007</i>	<i>-0.014</i>	<i>0.079</i>	<i>0.042</i>	<i>1.014</i>
Natural hedge	17,140	0.669	0.790	0.330	0.311	1.000
		<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>
Profitability	17,140	0.041	0.076	0.186	0.083	2.618
		<i>0.026</i>	<i>0.029</i>	<i>0.175</i>	<i>0.099</i>	<i>2.819</i>
Size (log of assets)	17,140	4.406	4.239	2.081	2.115	12.280
		<i>0.243</i>	<i>0.106</i>	<i>1.704</i>	<i>1.617</i>	<i>13.317</i>
Concentrated industries						
Leverage	1630	0.275 ^a	0.256 ^a	0.182 ^a	0.169 ^a	0.955
		<i>-0.005</i>	<i>-0.025</i>	<i>0.171^c</i>	<i>0.163^c</i>	<i>1.077</i>
Capital/labor	1630	0.048 ^a	0.027 ^a	0.070 ^a	0.027 ^c	1.170
		<i>-0.003</i>	<i>-0.005</i>	<i>0.058^a</i>	<i>0.017^c</i>	<i>1.180</i>
Risk	1630	0.074 ^a	0.049 ^a	0.081 ^a	0.038 ^a	0.632
		<i>-0.009</i>	<i>-0.010</i>	<i>0.071^a</i>	<i>0.028^a</i>	<i>0.795</i>
Natural hedge	1630	0.665 ^a	0.810 ^b	0.350 ^a	0.352 ^a	1.000
		<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>
Profitability	1630	0.072 ^a	0.085 ^a	0.135 ^a	0.066 ^a	1.940
		<i>0.015</i>	<i>0.016</i>	<i>0.128^a</i>	<i>0.081^a</i>	<i>1.737</i>
Size (log of assets)	1630	5.826 ^a	5.599 ^a	2.645 ^a	3.051 ^a	11.777
		<i>0.374</i>	<i>0.159</i>	<i>2.147^a</i>	<i>2.094^a</i>	<i>14.929</i>

Summary statistics for manufacturing firms operating in 315 competitive industries and 46 concentrated industries between 1981 and 2000. Superscripts show tests of differences between competitive and concentrated industries where a, b, c denote statistical significance at the 1, 5, and 10% confidence levels. Following the U.S. Department of Justice, we define competitive industries as those with a Herfindahl–Hirschman Index under 1000 and concentrated industries as those with a Herfindahl–Hirschman Index over 1800. Values in *italics* (the second row for each variable) are adjusted for industry-year. Robust dispersion is a nonparametric measure of dispersion, calculated as the range between the 69th and 31st percentiles, which is comparable to one standard deviation under normality. Leverage is total debt divided by total assets (book values). Capital/labor is net property, plant and equipment (in millions of dollars) per employee. Risk is the standard deviation of operating cash flow divided by assets. Natural hedge is one minus the absolute value of the difference between the firm's capital–labor ratio and the market-share weighted industry-year median (own firm excluded), normalized by the industry-year range. We do not demean natural hedge because it is industry-year adjusted by construction. Profitability is EBIT divided by assets.

5.2 Table 1: summary statistics

- **Read:** concentrated industries have **higher leverage**, **higher profitability**, and **larger firms**, while competitive industries have more dispersion in leverage and risk.

- Mean leverage is about **0.239** in competitive industries and **0.275** in concentrated industries.
- Median leverage is about **0.212** versus **0.256**.
- The paper interprets this as consistent with stronger strategic or collusive forces in concentrated industries.

5.3 Table 2: entrants, incumbents, and exiters

- **Read:** entrants begin with **high leverage**, **low capital intensity**, and **low profitability**; exiters are heavily leveraged, risky, and unprofitable.
- This supports the idea of an industry **core** and **fringe**.
- But it does *not* support the dynamic prediction that entrants should begin with lower debt and then lever up gradually.

Table 2
Subsample means for entrants, incumbents, and exiters

	Entrants	Post Entry	Incumbents	Pre Exit	Exiters
Competitive industries					
Leverage	0.296 <i>0.033</i>	0.218 ^a <i>0.002^b</i>	0.229 ^{a,c,a,a} <i>-0.013^{a,b,a,a}</i>	0.413 ^c <i>0.119^c</i>	0.470 ^a <i>0.167^a</i>
Capital/labor	0.031 <i>-0.014</i>	0.046 ^a <i>-0.010^c</i>	0.050 ^{a,c,b,b} <i>-0.003^{b,c,-,-}</i>	0.028 ^c <i>-0.008^c</i>	0.025 ^c <i>-0.001^b</i>
Risk	<i>n/a</i> <i>n/a</i>	0.133 ^{a/a} <i>0.002^{a/a}</i>	0.086 ^{a/a,a,a,a} <i>-0.010^{a/a,a,a,a}</i>	0.122 ^c <i>0.027^c</i>	0.126 ^{a/a} <i>0.025^{a/a}</i>
Natural hedge	0.753 <i>n/a</i>	0.750 ^c <i>n/a</i>	0.662 ^{a,a,-,-} <i>n/a</i>	0.680 ^c <i>n/a</i>	0.656 ^c <i>n/a</i>
Profitability	0.017 <i>0.002</i>	-0.052 ^a <i>0.003^c</i>	0.049 ^{a,a,a,a} <i>0.032^{a,a,a,a}</i>	-0.055 ^a <i>-0.072^a</i>	-0.153 ^a <i>-0.155^a</i>
Size (log of assets)	3.025 <i>-0.515</i>	4.069 ^a <i>0.173^a</i>	4.558 ^{a,a,a,a} <i>0.304^{a,b,a,a}</i>	3.032 ^c <i>-0.596^c</i>	2.779 ^c <i>-0.959^b</i>
Observations (firm-years)	349	636	13,661	95	44
Concentrated industries					
Leverage	0.313 <i>0.031</i>	0.316 ^c <i>0.017^c</i>	0.261 ^{c,c,b,b} <i>-0.017^{c,-,b,c}</i>	0.466 ^c <i>0.156^c</i>	0.731 ^b <i>0.375^c</i>
Capital/labor	0.031 <i>-0.004</i>	0.047 ^c <i>-0.023^c</i>	0.051 ^{b,c,c,c} <i>-0.003^{b,-,c,-,-}</i>	0.022 ^c <i>-0.005^c</i>	0.017 ^c <i>-0.004^c</i>
Risk	<i>n/a</i> <i>n/a</i>	0.083 ^{a/a} <i>0.005^{a/a}</i>	0.069 ^{a/a,c,-,-} <i>-0.011^{a/a,c,-,-}</i>	0.080 ^c <i>-0.012^c</i>	0.094 ^{a/a} <i>0.014^{a/a}</i>
Natural hedge	0.736 <i>n/a</i>	0.767 ^c <i>n/a</i>	0.650 ^{b,-,c,-,-} <i>n/a</i>	0.777 ^c <i>n/a</i>	0.948 ^c <i>n/a</i>
Profitability	0.027 <i>-0.006</i>	0.037 ^c <i>-0.015^c</i>	0.081 ^{c,b,a,-,-} <i>0.021^{c,c,c,-,-}</i>	-0.041 ^c <i>-0.042^c</i>	-0.023 ^c <i>-0.058^c</i>
Size (log of assets)	3.140 <i>-1.947</i>	4.766 ^a <i>-0.553^a</i>	6.067 ^{a,a,a,-,-} <i>0.500^{a,a,-,-}</i>	3.800 ^c <i>0.028^c</i>	3.876 ^c <i>-0.689^c</i>
Observations (firm-years)	19	48	1,319	8	2

Subsample means for manufacturing firms operating in 315 competitive industries and 46 concentrated industries between 1981 and 2000. Superscripts in column 2 show tests of differences in means between entrants and post entry. Superscripts in column 3 show tests of differences between incumbents and all other groups. Superscripts in column 4 show tests of differences between pre exit and exiters, and in column 5 between entrants and exiters—where a, b, and c denote statistical significance at the 1, 5, and 10% confidence levels. Following the U.S. Department of Justice, we define competitive industries as those with a Herfindahl-Hirschman Index over 1000 and concentrated industries as those with a Herfindahl-Hirschman Index over 1800. Values in *italics* are adjusted for industry-year. Entrants are firms that enter the panel in the last ten years of the sample period (1991–2000). Exiters are firms that leave the panel in 1991–2000 through bankruptcy (Chapter 11) or liquidation (Chapter 7). Incumbents are firms present in both subperiods, though not necessarily in every year of the panel. Means for entrants (exiters) pertain to their first (last) year in the sample. Means for post-entry (pre-exit) pertain to the years after (before) entry (exit). Leverage is total debt divided by total assets (book values). Capital/labor is net property, plant and equipment (in millions of dollars) per employee. Risk is the standard deviation of operating cash flow divided by assets. Natural hedge is one minus the absolute value of the difference between the firm's capital-labor ratio and the market-share weighted industry-year median (own firm excluded), normalized by the industry-year range. We do not demean natural hedge because it is industry-year adjusted by construction. Profitability is EBIT divided by assets.

Table 3
Industry effects and reversion in financial structure for competitive industries

Industry financial structure				Adjusted R ²		
	2-SIC	3-SIC	4-SIC	Adjusted R ²	with firm fixed effects	Firm-years
A. Importance of industry levels						
Leverage	0.048 (1.37)	0.013 (0.54)	0.186 (12.40) ^a	13.0%	67.0%	15,719
Capital/Labor	0.350 (14.00) ^a	0.198 (6.83) ^a	0.158 (7.90) ^a	45.0%	85.0%	15,707
Risk	0.068 (1.84) ^c	0.003 (0.13)	0.141 (10.07) ^a	3.8%	87.6%	15,553
B. Importance of industry changes						
ΔLeverage	0.026 (0.81)	0.076 (4.22) ^a	0.277 (25.65) ^a	5.0%		15,719
ΔCapital/labor	-0.002 (-0.08)	0.031 (1.82) ^c	0.119 (9.92) ^a	1.0%		15,707
ΔRisk	-0.012 (-0.41)	0.059 (4.21) ^a	0.038 (4.75) ^a	0.4%		15,553
C. Reversion to industry levels						
ΔLeverage	-0.032 (-2.91) ^a	-0.031 (-2.07) ^b	-0.104 (-8.00) ^a	8.0%		15,719
ΔCapital/Labor	0.030 (1.88) ^c	-0.045 (-3.46) ^a	-0.035 (-3.89) ^a	1.0%		15,707
ΔRisk	-0.006 (-0.75)	-0.013 (-1.39)	-0.041 (-5.91) ^a	4.0%		15,553

Ordinary least squares regressions of firm-level variables on industry-level variables for firms in competitively-structured industries between 1981 and 2000. (*t*-statistics in parentheses). a, b, c denote statistical significance at the 1, 5, and 10% confidence levels. Leverage is total debt divided by total assets (book values). Capital/labor is net property, plant and equipment (in millions of dollars) per employee. Risk is the standard deviation of operating cash flow divided by assets. Panel A regresses the firm-level variables on lagged industry-year medians. Panel B regresses changes in firm-level variables on concurrent changes in industry-year medians. For leverage we estimate: $\Delta \text{firm leverage}_{i,t-j} = \beta' \Delta \text{industry median leverage}_{i,t-j}$. Panel C regresses changes in firm-level variables on lagged deviations from industry-year medians - for leverage we estimate: $\Delta \text{firm leverage}_{i,t-j} = \beta' (\text{firm leverage}_{i,t-j} - \text{industry medians leverage}_{i,t-j})$. Industry medians exclude the firm itself, and the three (two)-digit SIC medians exclude firms in the finer four (three)-digit SIC industries.

5.4 Table 3: industry effects and slow reversion

- **Panel A:** industry levels explain only **13%** of leverage variation, but **45%** of capital-labor variation.
- **Panel B:** firms react only weakly to industry-wide changes.
- **Panel C:** firms revert to industry medians slowly; annual leverage reversion is only about **3%–10%**, depending on how industry is defined.

- Main lesson: firms are persistent in their relative industry positions.

5.5 Table 4: transition frequencies

Table 4
Transition frequencies for competitive industries

	Firm status in 1991–2000							
	Q1	Q2	Q3	Q4	Q5	Exit	Sell	Total
Leverage								
Enter	25	18	18	20	18 ^b			17 ^c
Switch	12	19	21	25	23 ^a			11 ^b
Q1	32 ^a		9	4	2	3	32	10 ^a
Q2	40	28 ^a	19	9	3	1	30	18 ^a
Q3	4	17	22 ^a	18	7	2	32	17 ^a
Q4	2	9	17	23 ^a	15	4	31	17 ^a
Q5	1	5	7	17	30 ^a	9 ^a	31 ^c	12 ^a
Total	11	17	17	17	13	3 ^a	22 ^a	100
Capital/Labor								
Enter	14	22	22	21	22 ^b			17 ^c
Switch	19	25	23	15	18 ^c			11 ^c
Q1	30 ^a	21	5	0	0	8	36	10 ^a
Q2	12	27 ^a	15	7	1	3	35	17 ^a
Q3	5	19	21 ^a	14	5	3	33	16 ^a
Q4	2	8	18	29 ^a	13	2	27	17 ^a
Q5	2	2	11	28	32 ^a	3 ^c	23 ^c	12 ^a
Total	11	18	17	17	13	3 ^c	22 ^a	100
Risk								
Enter	18	19	20	23	20 ^c			17 ^c
Switch	13	21	21	27	18 ^c			11 ^b
Q1	30 ^a	27	11	5	0	0	27	10 ^a
Q2	16	29 ^a	17	8	2	1	27	17 ^a
Q3	6	18	19 ^a	18	3	4	32	17 ^a
Q4	3	9	18	24 ^a	8	5	33	17 ^a
Q5	0	3	10	16	28 ^a	8 ^a	35 ^b	11 ^a
Total	12	18	17	17	11	3 ^a	22 ^a	100
Natural hedge								
Enter	16	20	19	23	22 ^c			17 ^c
Switch	18	15	19	25	22 ^c			11 ^c
Q1	33 ^a	23	8	1	2	5	27	10 ^a
Q2	12	30 ^a	16	8	2	3	30	17 ^a
Q3	5	18	24 ^a	15	4	2	32	16 ^a
Q4	2	7	16	28 ^a	12	3	32	17 ^a
Q5	0	1	8	20	31 ^a	6 ^c	33 ^b	12 ^a
Total	11	17	17	18	13	3 ^c	22 ^a	100
Profitability								
Enter	21	21	20	20	18 ^c			17 ^c
Switch	17	24	20	24	15 ^c			11 ^c
Q1	17 ^a	19	6	6	3	12	36	10 ^a
Q2	12	23 ^a	16	11	5	4	29	17 ^a
Q3	6	17	22 ^a	16	7	2	30	17 ^a
Q4	2	10	18	23 ^a	14	2	31	17 ^a
Q5	2	7	14	18	28 ^a	0 ^a	31 ^c	12 ^a
Total	11	17	17	17	13	3 ^a	22 ^b	100
Size (log of assets)								
Enter	12	17	25	28	18 ^b			17 ^a
Switch	19	30	19	16	16 ^c			11 ^a
Q1	36 ^a	18	3	0	0	8	34	9 ^a
Q2	13	26 ^a	15	3	1	4	38	17 ^a

Table 4
(continued)

	Firm status in 1991–2000							Total
	Q1	Q2	Q3	Q4	Q5	Exit	Sell	
Q3	5	19	26 ^a	10	2	3	35	17 ^a
Q4	1	7	19	35 ^a	11	2	25	18 ^a
Q5	0	1	3	22	51 ^a	2 ^a	22 ^c	12 ^a
Total	11	17	17	17	13	3 ^c	22 ^a	100

Quintile transition frequencies for firm-period means between 1981–1990 and 1991–2000 for firms that enter or switch industries, incumbents, and firms that leave the panel through exit or sale. Each panel shows percentages relative to row totals. Superscripts in the last column and bottom row of each panel show Chi-square tests that industry demographics follow a uniform distribution. Superscripts in quintile five test whether the percentages in quintiles one and five are equal. Superscripts on the diagonal quintiles test whether adjacent quintiles are equal. a, b, c denote statistical significance at the 1, 5, and 10% confidence levels. Leverage is total debt divided by total assets (book values). Capital/labor is net property, plant and equipment (in millions of dollars) per employee. Risk is the standard deviation of operating cash flow divided by assets. Natural hedge is one minus the absolute value of the difference between the firm's capital-labor ratio and the market-share weighted industry-year median (own firm excluded), normalized by the industry-year range. Profitability is EBIT divided by assets.

- **Read:** large, capital-intensive, profitable firms stay near the top of their industry distributions, while small firms churn.
- The paper interprets this as evidence that industry equilibrium may **sustain diversity** rather than eliminate it.
- The high exit and sell-out rates among the smallest firms reinforce the core-fringe interpretation.

5.6 Table 5: OLS versus GMM simultaneous equations

- **Read:** once the paper controls properly for simultaneity, several relations change sign.
- The natural-hedge effect on leverage remains negative across estimation methods, which is one of the paper's main robust findings.
- Intra- and extra-quintile peer changes enter negatively and significantly in all three equations.
- The authors therefore prefer **GMM** to **OLS** for economic interpretation.

Table 5
Ordinary least squares and generalized method of moments regressions for competitive industries

Dependent variables	Ordinary least squares			General method of moments		
	Leverage	K/L	Risk	Leverage	K/L	Risk
Leverage		0.014 (5.83) ^a	-0.001 (-0.44)		0.005 (0.27)	0.196 (5.56) ^a
Capital/Labor	0.139 (6.27) ^a		-0.002 (-0.48)	-1.049 (-3.62) ^a		0.218 (2.76) ^a
Risk	-0.019 (-0.51)	-0.002 (-0.17)		2.786 (9.89) ^a	0.034 (0.86)	
<i>Industry Variables</i>						
Natural hedge	-0.021 (-5.74) ^a	-0.025 (-21.00) ^a	0.000 (-0.58)	-0.727 (-7.14) ^a	-0.004 (-0.26)	0.153 (4.52) ^a
(Natural hedge) ²	0.013 (2.38) ^a	0.001 (0.39)	-0.004 (-3.39) ^a	0.917 (8.40) ^a	0.010 (0.50)	-0.201 (-5.99) ^a
Intra-quantile change	-0.303 (-21.23) ^a	-0.434 (-40.17) ^a	-0.498 (-31.08) ^a	-0.105 (-2.75) ^a	-0.419 (-5.06) ^a	-0.246 (-3.04) ^a
Extra-quantile change	-0.023 (-36.40) ^a	-0.071 (-36.77) ^a	-0.040 (-33.86) ^a	-0.006 (-2.57) ^a	-0.060 (-5.57) ^a	-0.015 (-2.56) ^a
<i>Control Variables</i>						
Profitability	-0.163 (-26.30) ^a	-0.001 (-0.71)	-0.046 (-36.88) ^a	-0.054 (-0.34)	0.009 (0.66)	0.060 (1.05)
(Profitability) ²	0.072 (7.84) ^a	-0.007 (-2.15) ^a	0.087 (49.50) ^a	0.440 (2.76) ^a	0.005 (0.30)	-0.107 (-2.05) ^a
Size (log of assets)	0.001 (25.82) ^a	0.005 (4.70) ^a	-0.005 (-7.20) ^a	1.146 (9.42) ^a	0.013 (0.58)	-0.242 (-5.24) ^a
Diversification	0.013 (1.36)	0.001 (0.44)	0.004 (2.45) ^a	-0.874 (-4.02) ^a	0.027 (1.16)	0.175 (2.71) ^a
Tobin's q	-0.005 (-5.94) ^a	-0.001 (-4.16) ^a	0.002 (0.09) ^a	0.271 (6.41) ^a	-0.001 (-0.17)	-0.061 (-3.58) ^a
<i>Fixed Effects</i>						
Year	0.005 (1.24)	-0.001 (-0.97)	0.000 (0.32)	-0.003 (-0.10)	-0.002 (-1.62)	0.002 (0.42)
Switcher	0.009 (3.77) ^a	-0.001 (-0.63)	-0.003 (-5.71) ^a	0.021 (1.89) ^a	-0.001 (-0.82)	-0.004 (-1.29)
Exit	0.033 (1.87) ^a	-0.004 (-1.48)	-0.004 (-2.07) ^a	0.053 (1.71) ^a	-0.003 (-0.74)	-0.008 (-1.94)
Seller	0.004 (2.20) ^a	-0.001 (-0.73)	-0.002 (-3.88) ^a	-0.021 (-1.98) ^a	-0.001 (-1.07)	0.006 (1.86) ^a
Adjusted R ²	12.54%	12.89%	28.16%	n/a	n/a	n/a
Degrees of freedom	17,125	17,125	17,125	17,125	17,125	17,125
Wald test, H ₀ : all coefficients = 0	2.482 ^a	2.551 ^a	6.756 ^a	191 ^a	62 ^a	54 ^a
Hauser-Jacobson (10 def)	n/a	n/a	n/a	7	10 ^a	2 ^a
Instrument relevance	n/a	n/a	n/a	8.49%	9.60%	17.78%

The continuous variables are first-differenced to control for firm fixed effects. The regressions also control for four-digit SIC industry-year fixed effects. For our GMM regressions, we instrument all variables except for the entry/exit dummy variables and quantile changes by their second lags in levels. We also include the four dummy variables, current and lagged quantile changes (twelve variables), and a constant term in the instrument set, resulting in ten one-identifying restrictions per equation. Leverage is total debt divided by total assets (book value). Capital/labor is net property, plant and equipment (in millions of dollars) per employee. Risk is the standard deviation of operating cash flow divided by assets. Natural hedge is one minus the absolute value of the difference between the firm's capital-labor ratio and the market-share weighted industry-year median (own firm excluded), normalized by the industry-year range. Intra-quantile change is the mean change in the dependent variable within the industry half to which a firm belongs (own firm excluded). Extra-quantile change is the mean change in the other half of the industry. These quantiles are based on the lagged levels of the dependent variables. Profitability is EBIT divided by assets. Diversification is one minus the Herfindahl of output across the firm's four-digit SIC industries. Tobin's q is the market value of equity plus the book value of debt and preferred stock minus deferred taxes, divided by the book value of assets. For GMM, we use the Newey-West procedure to address heteroskedasticity and the autocorrelation introduced by first-differencing. Asymptotic t-values in parentheses, a, b, c denote statistical significance at the 1, 5, and 10% confidence levels.

Table 6
Economic significance of the determinants of financial leverage, capital intensity, and risk

Dependent variables	25th percentile			50th percentile			75th percentile		
	Debt	K/L	Risk	Debt	K/L	Risk	Debt	K/L	Risk
Competitive industries									
Leverage	n/a		-1.32	n/a		0.25	n/a		1.68
Capital/labor	0.03	n/a	-0.05	-0.28	n/a	0.09	-0.60	n/a	0.22
Risk	-1.68		n/a	0.12		n/a	1.92		n/a
<i>Industry Variables</i>									
Natural hedge	3.37		-1.51	0.79		-0.38	-3.28		1.39
Intra-quantile change	0.34	0.08	0.40	-0.27	0.00	0.08	-0.87	-0.08	-0.26
Extra-quantile change	0.01	0.09	0.16	-0.23	0.00	0.07	-0.51	-0.09	-0.03
<i>Control Variables</i>									
Profitability	-0.14		-0.18	-0.26		0.10	-0.37		0.35
Size (log of assets)	-6.26		2.69	-0.61		0.23	5.46		-2.42
Diversification	1.07		-0.48	-0.15		0.03	-1.25		0.49
Tobin's q	-2.67		1.20	0.90		-0.46	4.03		-1.92
Concentrated industries									
Leverage	n/a		-0.26	n/a		0.01	n/a		0.27
Capital/labor	n/a		n/a	n/a		n/a	n/a		n/a
Risk	-0.40		n/a	0.01		n/a	0.45		n/a
<i>Industry Variables</i>									
Natural hedge									
Intra-quantile change	2.39	0.29	0.91	-0.08	-0.01	0.03	-2.38	-0.26	-0.92
Extra-quantile change	2.06	0.16	0.97	-0.05	0.01	0.02	-2.25	-0.18	-0.98
<i>Control Variables</i>									
Profitability			-0.24			0.01			0.25
Size (log of assets)	-1.13		0.09			0.00	1.23		-0.06
Diversification			0.05			-0.06			-0.50
Tobin's q			0.40						

Predicted changes in the dependent variables (using generalized method of moments estimates) as each regressor varies from the 25th to 75th sample percentile, holding all other regressors at their sample mean levels (statistically insignificant relations omitted). Reported are percentages relative to the sample mean and normalized by the sample range (maximum minus minimum). Leverage is total debt divided by total assets. Capital/labor is net property, plant and equipment (in millions of dollars) per employee. Risk is the standard deviation of operating cash flow divided by sales. Natural hedge is one minus the absolute value of the difference between the firm's capital-labor ratio and the market-share weighted industry-year median (own firm excluded), normalized by the industry-year range. Intra-quantile change is the mean change in the dependent variable within the industry half to which a firm belongs (own firm excluded). Extra-quantile change is the mean change in the other half of the industry. These quantiles are based on the lagged levels of the dependent variables. Profitability is EBIT divided by assets. Diversification is one minus the Herfindahl of output across the firm's four-digit SIC industries. Tobin's q is the market value of equity plus the book value of debt and preferred stock minus deferred taxes, divided by the book value of assets.

5.7 Table 6: economic significance

- **Read:** natural hedge is not only statistically significant but also economically meaningful in competitive industries.
- Moving from the 50th to the 25th or 75th natural-hedge percentile changes leverage by about **3.3%** in either direction, which the authors describe as a difference of **well over six percentage points** between tails.
- Peer-change variables are statistically strong but economically modest in competitive industries—about **one percentage point** for leverage.
- In concentrated industries, peer interactions are much stronger: moving across the intra- and extra-quantile distributions shifts leverage by **more than four percentage points**.

6 Short summary

- **Method.** Use a panel of firms, construct within-industry position measures, and estimate leverage, capital intensity, and risk jointly with first-differenced GMM.
- **Identification.** Remove firm effects by differencing, isolate within-industry variation with industry-year effects, and instrument endogenous regressors with second lags.
- **Main finding.** Industry matters far more through **within-industry position** than through crude industry dummies.
- **Main result.** In competitive industries, firms near the industry median technology use less debt and adjust against their peers; in concentrated industries, leverage is higher and peer interactions

are stronger, while the natural-hedge effect disappears.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that industry affects capital structure in a richer way than standard industry fixed effects suggest.
- Most leverage variation is inside industries, and part of that variation is systematically related to a firm's technological and strategic position inside the industry.
- Real and financial decisions must be analyzed jointly.

7.2 Economic intuition

- A firm that sits near the industry's technological center is naturally hedged and has less need for debt.
- Firms do not move in isolation. Their leverage, technology, and risk choices respond to what neighboring firms in the industry are doing.
- Competitive industries do not force firms into identical positions. Instead, equilibrium can sustain a diverse mix of core and fringe firms.

7.3 Takeaway

This paper reframes capital structure as an **industry equilibrium outcome** rather than a purely firm-specific financing choice. The main lesson is simple: to understand why firms borrow differently, one must look not only across industries, but also **within** them—at technology, peer behavior, and the joint determination of leverage and risk.